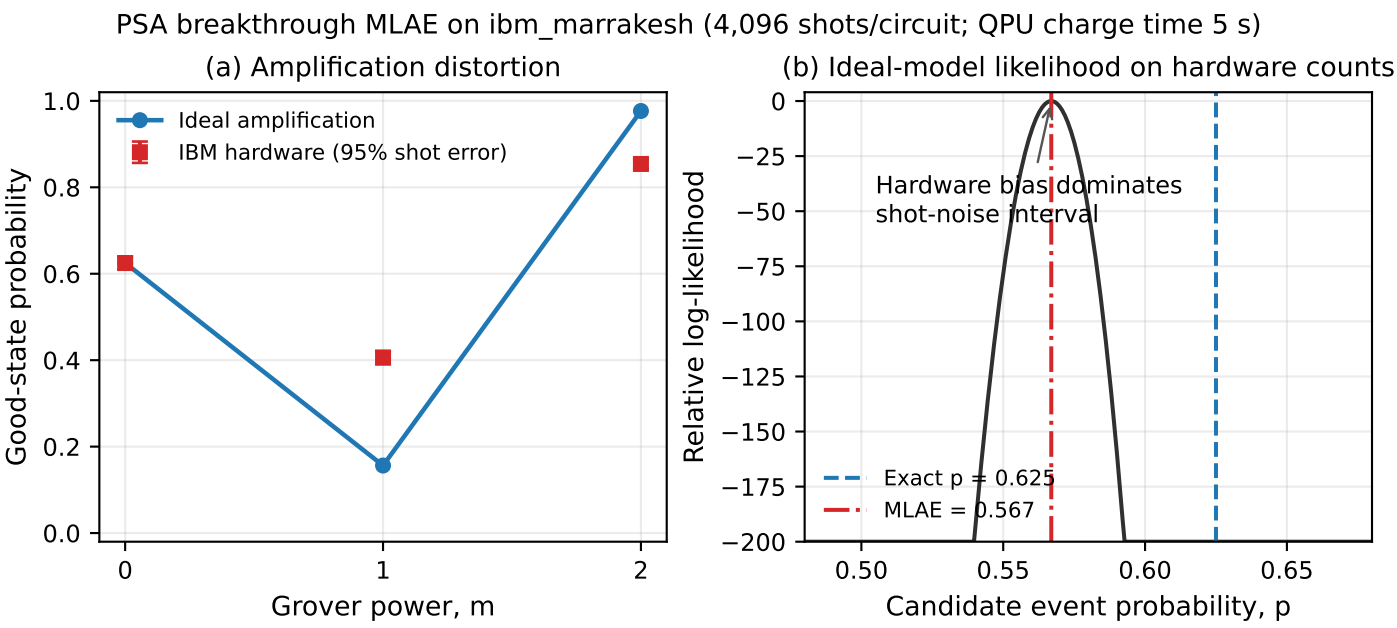
PSA breakthrough MLAE on ibm_marrakesh (4,096 shots/circuit; QPU charge time 5 s)
(a) Amplification distortion
Ideal amplification
IBM hardware (95% shot error)
Good-state probability
Grover power, m
(b) Ideal-model likelihood on hardware counts
Hardware bias dominates
shot-noise interval
Exact p = 0.625
MLAE = 0.567
Relative log-likelihood
Candidate event probability, p